

ENME485/691 Industrial AI
Homework 02 - Fall 2026
Due: Sep 28 (before 12:30 pm)

Objective: Implement the analytical tools and assess the health of the shaft in a rotor-bearing system.

1 Background

A rotor-bearing testbed was built to analyze the health condition of the shaft with unbalance defects.

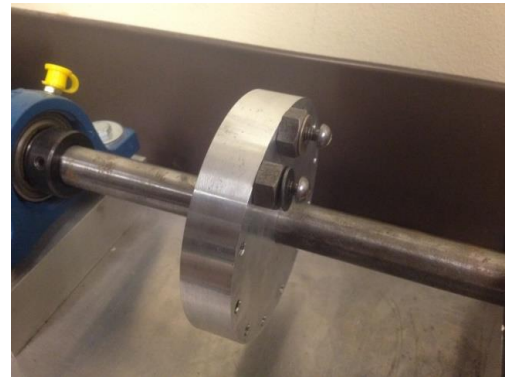


Fig. 1 Testbed

The rotating speed of the test-bed is **20 Hz**, and screws can be added on the disk of the shaft to induce unbalance defects. An accelerometer was mounted on the bearing block and the vibrations could be measured. The vibration data were collected under normal condition and two unbalanced conditions (1 screw on the disk & 2 screws on the disk). The sampling rate of data collection is **2560 Hz**.



(a)



(b)

Fig. 2 Shaft unbalance: (a) unbalance level 1 and (b) unbalance level 2

2 Problem Statement

Develop a logistic regression method for assessing the health condition of the shaft based on provided vibration data.

3 Data Source

Two folders will be uploaded. Use the following files from each folder for your assessment.

Training Data:

Healthy – vibration data measured under normal condition (20 observations).

Faulty – vibration data measured under shaft unbalance condition level 2 (2 screws on the disk) (20 observations).

Testing Data:

Vibration data measured from the testbed with unknown health conditions (30 observations). All health assessment results should come from testing data sets.

4 Suggested Algorithms

Feature Extraction	The amplitude of 1X harmonic component in frequency domain (as well as other features of your choosing)
Modelling	Logistic Regression

5 Result Submission

For each **group leader**, please submit a (five to ten minute) presentation file and a report document presenting the health assessment result. The submission report should contain the following sections:

- (1) Introduction
 - a. background information and description of the problem
 - b. raw data description
- (2) Implemented Solution
 - a. brief introduction of the analytics process (sample preparation, feature extraction and selection, application of suggested algorithms)
 - b. flow chart of entire solution process
- (3) Results
 - a. figures or tables showing the assessment results, including the health label of each test sample
- (4) Discussion
 - a. include any observations about the modelling techniques, extracted features, results, etc.
- (5) Conclusion
 - a. brief statement concluding the important findings
- (6) Appendix – Codes
 - a. all the programming codes used to achieve the results in the report